

Mechanistic insights into de-inking

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Abstract

Mechanistic parallels between de-inking and laundering are complicated by the greater role of mechanical forces in de-inking processes. However, laboratory experiments indicate that ink removal from ledger paper, newspaper and a newspaper/magazine mixture is related to the de-inking surfactant cloud point — a clear parallel with detergency. This suggests there are similarities in the mechanisms of ink and soil detachment from fibers. Both the rollback mechanism and emulsification probably occur in the removal of liquid inks from cellulose fibers. The importance of solubilization in detaching ink from cellulose is less clear. This mechanism may compete with ink detachment promoted by the mechanical disaggregation of fibers. The softening of solid inks facilitates the detachment from cellulose by hydrodynamic forces. This process may well dominate the surfactant-promoted wetting and adsorption effects thought to operate in solid soil removal from cloth. Ink softening promotes the agglomeration of dispersed ink particles when they collide.

Keywords: De-inking; Emulsification; Laundering; Solubilization; Surfactants

1. Introduction

The British Patent Office granted the first de-inking patent in 1800. However, de-inking has been practised on a large commercial scale only since World War II. The convergent evolution of North American, European and Japanese de-inking processes in the last 10 years has been discussed [1]. Various workers have noted that de-inking is fundamentally a laundering process [2,3]. The adhesion of ink to cellulose and soils to cloth is due to van der Waals, electrical and mechanical forces. In both de-inking and laundering, a stain is removed from fibers by a combination of chemical and mechanical mechanisms. Like laundering, de-inking is a two-step process: detachment of the stain from the fiber and separation of the dispersed staining material from the fiber. This last step is more difficult in de-inking than in

laundering because the paper has been disaggregated into individual fibers.

Few mechanistic studies of de-inking have been reported. The objective of this paper is to review the literature to examine parallels between de-inking and laundering. Some original results will also be presented. An effort will be made to determine whether the de-inking results are consistent with the surfactant behavior observed in detergency studies.

1.1. Repulping used paper

In de-inking, used paper is first repulped — disaggregated into individual fibers under higher shear conditions than are used in laundering. Typical pulping conditions for old paper include a moderately high temperature, a fairly high pH, and high agitation. Process temperatures are in the

Table 1
Comparison of de-inking and laundering

Parameter	Laundering	De-inking
Substrate	Cloth fibers	Paper fibers
Soil	Food stains, sweat, body oils, blood, grass stains	Various inks
Cleaning medium	Surfactants + builder + sequestrant in water	Surfactant + pulping chemicals ^a in water
Process temperature (°F)	70–150	70–200
pH	8–10	9–12

^aSee Table 2 for a summary of common pulping chemicals.

range 37–93°C (100–200°F). The pH is usually in the range 9–12. Typical pulping conditions for old newspapers and for office papers are given in Table 1 and are compared to representative U.S. laundering conditions.

For de-inking, surfactant is usually present in the pulper. Additional chemicals are often used (Table 2). These include hydrogen peroxide bleach to limit yellowing of newsprint caused by the effect

of high pH on lignin. Complexing agents are used to protect the hydrogen peroxide from reaction with iron and other metal ions that may be present in the process water. Sodium silicate is often used and provides a high pH, possibly some detergent action and some complexing ability.

The mechanical energy inputs are much greater in de-inking than in laundering. This can be seen easily by looking inside a pulper and a home or

Table 2
De-inking process chemicals

Chemical	Addition point	Function
Surfactant	Pulper	(1) Promotes ink detachment from cellulose fibers. (2) Disperses ink into small particles for removal by washing. (3) For flotation, renders ink particles hydrophobic promoting ink attachment to air bubbles. Stabilizes foam. May also add surfactant at the flotation stage for this purpose
Sodium hydroxide	Pulper	Raises pH to 8–10 to promote fiber swelling which aids ink removal and disintegration of paper into individual fibers.
Sodium silicate	Pulper	(1) Dispersant for detached ink particles. (2) Raises pH. (3) complex multivalent metal ions.
Hydrogen peroxide	Bleach tower and/or pulper	(1) Prevents lignin-yellowing of pulp in the pulper. (2) Bleach tower — increases pulp brightness. ^a
Complexing agent	Bleach tower ^b and/or pulper	Stabilizes hydrogen peroxide against attack by oxidizable metal ions. DTPA ^c and EDTA ^d are most commonly used.
Fatty acid	Flotation cell	Used in combination with a soluble calcium salt, most commonly calcium chloride, to generate a calcium salt in situ. Renders ink particles hydrophobic and stabilizes foam.

^aOther oxidative bleaching agents that may be used in bleach towers include sodium hypochlorite and chlorine dioxide. Reductive bleaches include sodium hydrosulfite and formamidine sulfinic acid.

^bIn conjunction with a peroxygen bleach, particularly hydrogen peroxide.

^cDiethylenepentaminetetraacetic acid.

^dEthylenediaminetetraacetic acid.

commercial laundry washing machine. Mechanical forces probably play a greater role in de-inking than in laundering. During pulping, mechanical forces promote ink detachment from cellulose fibers by

- (1) hydrodynamic flow of the liquid phase;
- (2) fiber swelling;
- (3) fiber flexing;
- (4) fiber–fiber abrasion.

Purely mechanical processes called high temperature dispersion [4–9] and kneading [10–13] are sometimes employed after repulping the old paper. In these steps, interfiber abrasion promotes the detachment of additional ink from fibers. These processes also break down large dispersed ink particles into smaller ones more easily removed by flotation and by washing (see below). Centrifugal cleaners can remove dispersed particles denser than water (forward cleaners) and lighter than water (reverse or throughflow cleaners). Reverse cleaners are primarily intended to remove adhesive particles. Forward cleaners are useful in removing larger ink particles such as those produced when de-inking laser-printed and photocopy paper (Fig. 1). Screens primarily remove contaminants larger than ink particles (Fig. 1). Surfactants and other chemicals can be useful in adjusting the density of these larger ink particles for easier removal in cleaners (see below).

1.2. Washing and flotation

The primary processes used to separate dispersed ink from cellulose fibers are washing and flotation. In washing, water containing dispersed ink is drained from the cellulose fibers. Clean water is added, the cellulose fibers are dispersed. This pro-

cess is repeated one or more times. The analogy to laundering is obvious.

Flotation is a chemical–mechanical process with no analogy to laundering. Air is rapidly bubbled through the pulp slurry. Some surfactants promote foam formation and the attachment of ink particles to the bubble surfaces. This is considered to be an agglomeration process. The bubbles rise to the top of the aqueous pulp slurry forming a foam layer at the surface. The ink particles are trapped in this froth layer. Commercial flotation cells use various methods to separate the ink-laden froth from the pulp slurry. This froth layer may be removed by flowing over the top of the flotation cell, being scraped off the surface of the pulp slurry, or being forced out of the flotation cell by pressurized air. The foam must have enough stability to be removed before many of the bubbles can break and return ink to the cellulose fiber slurry.

The total ink particle size range of flotation effectiveness is 10–100 μm (Fig. 1) [14,15]. Greatest effectiveness is achieved in the 30–80 μm range [14,15]. The mechanism of bubble–ink particle attachment has been the subject of computer modeling studies [16].

Washing is most efficient over an ink particle diameter range of 1–25 μm (Fig. 1), with the highest effectiveness in the 5–15 μm range [14,15]. Washing is most effective with dispersible inks and when the efficient removal of paper fillers and coatings is needed (as when manufacturing tissue products from used paper).

1.3. Ink chemistry

Ink chemistry and ink particle size after repulping will determine the relative effectiveness of different de-inking surfactants and which processes are most effective in separating the dispersed ink from the cellulose fibers (Table 3). References 3 and 16–19 describe ink chemistry.

Environmental scanning electron microscopy (ESEM) studies indicate that many liquid-based inks exist as films on cellulose fibers [21,22]. This can be seen in Fig. 2. Solid (toner) inks include xerographic and laser inks. ESEM micrographs (Fig. 3) clearly indicate their solid nature as par-

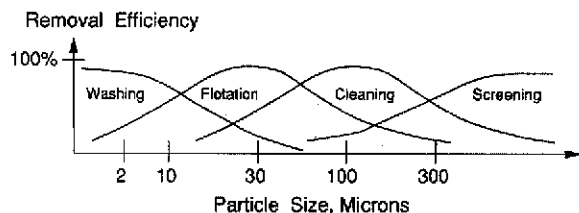


Fig. 1. Ink particle removal efficiency for different de-inking process units (adapted from Ref. 15).

Table 3

Ink particle size after repulping old paper (50)

Printing process	Ink particle size (μm)	
	Uncoated paper	Coated paper
Letterpress	2–30	10–100
Offset	2–30	5–100
Flexography	0.3–1	0.7–2
Glavure	2–30	5–30
Laser, xerographic	40–400	40–400

tially fused polymeric beads forming a thick layer on the paper surfaces [22].

2. Effect of process temperature on detergency and de-inking

The predominant surfactants used in wash de-inking old newspapers are non-ionic. Most of the proprietary surfactants used in the flotation de-inking of mixtures of old newspapers and magazines are non-ionic. (Fatty acid soaps generated in situ by the interaction of calcium ions with fatty

acids in the flotation cell are widely used, primarily in Europe.) De-inking results for non-ionic surfactants have been related to surfactant hydrophilic–lipophilic balance (HLB) values [23]. However, this approach neglects the effect of changes in the process temperature. Also, HLB values are difficult to calculate for many surfactant formulations containing more than one component. Various experimental methods for determining HLB values are known [24–30]. However, these procedures are lengthy and exacting. Surfactant cloud points are more easily measured. This makes it worthwhile to consider the de-inking process in terms of the surfactant cloud point.

The relationship of the non-ionic surfactant cloud point to the soil removal efficiency from cloth fibers has been studied extensively [31–34]. In laundering, optimum removal of non-polar oil phases from polyester/cotton fabrics occurs when the wash temperature is 15–25°C above the surfactant cloud point [35,36]. The optimum temperature for non-polar soil removal from fabrics using alcohol ethoxylates [37] and other non-ionic surfactants [38,39] occurs well above the surfactant cloud point at the phase inversion temperature



Fig. 2. Environmental scanning electron microscope image of newspaper print.

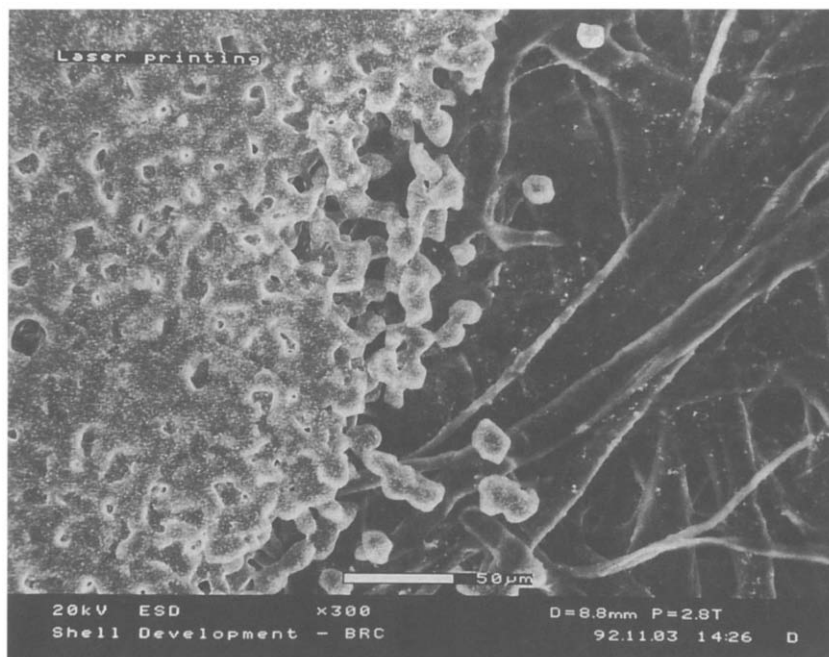


Fig. 3. Environmental scanning electron microscope image of laser-printed paper.

[33]. (The phase inversion temperature is defined as the temperature at which an oil-in-water emulsion inverts to a water-in-oil emulsion on heating.) In sharp contrast, the optimum removal of blends of non-polar and polar oil phases from fabric occurs at wash temperatures 0–30°C below the surfactant cloud point [37].

Figure 4 summarizes wash de-inking results for

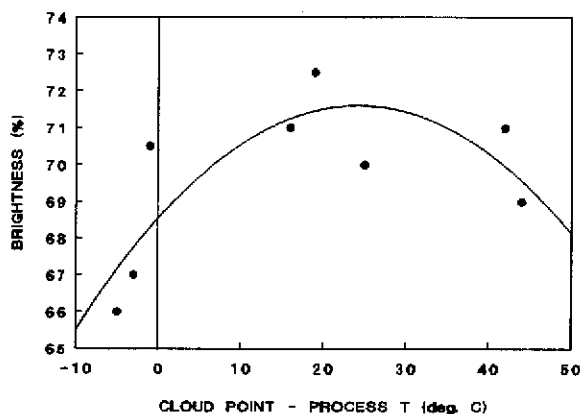


Fig. 4. Effect of surfactant cloud point on wash de-inking of white ledger paper.

a printed ledger paper, and indicates that maximum de-inking sheet brightness was obtained at process temperatures approximately 20–30°C below the cloud point. Thus, this ink appeared to behave as a polar oil, rather than a non-polar oil.

Newsprint wash de-inking data (Table 4 and Fig. 5) indicate that the highest de-inked sheet brightness values were obtained when the process temperature was approximately equal to the non-ionic surfactant cloud point [40,41]. The dirt count, a measure of residual ink particles on de-inked sheets, was at a minimum when the de-inking process temperature was approximately equal to the de-inking surfactant cloud point (Table 4). Figure 5 indicates that the de-inked newsprint/magazine brightness decreased when the process temperature was increased significantly above the surfactant cloud point. The temperature dependence of the de-inking efficiency is more consistent with these inks behaving like a polar oil than like a non-polar oil.

Why does the relationship of cloud point and optimum de-inking temperature differ for results summarized in Table 4 and Figs. 4 and 5?

Table 4

The relationship of de-inking process temperature and surfactant cloud point on ink removal^a

Test surfactant ^b	Surfactant cloud point (°C)	Process temperature (°C)	De-inked sheet	
			Brightness ^c (%)	Dirt count ^c (ppm)
C _{12–15} (EO) ₉	74	40	48–49	40–45
C _{12–15} (EO) ₉	74	50	49–50	40
C _{12–15} (EO) ₉	74	70	52	15
C _{14–15} (EO) ₇	40	40	49–51	20
C _{14–15} (EO) ₇	40	50	48	30
C _{14–15} (EO) ₇	40	70	48	40

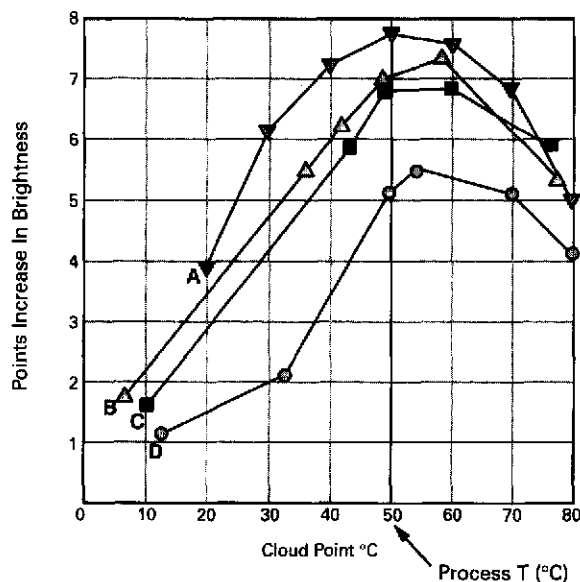
^aTaken from Ref. 40.^bSee text for key to abbreviations.^cThe brightness and dirt count measurements are described in Ref. 22.

Fig. 5. Effect of surfactant cloud point on wash de-inking of a 1:1 newsprint:magazine blend (see Ref. 33) (surfactant concentration 0.4% by weight relative to dry paper): curve A, blend of C₉–C₁₁ alcohol ethoxylate and C₁₂–C₁₅ alcohol ethoxylate; curve B, C₉–C₁₁ alcohol ethoxylate; curve C, C₁₂–C₁₅ alcohol ethoxylate; curve D nonylphenol ethoxylate.

Newspapers printed with a letterpress ink were used for the tests summarized in Table 4. The tests summarized in Fig. 5 used 50% by weight of newspapers printed with a letterpress ink and 50% by weight of magazines printed with an offset lithographic ink. The ledger ink (Fig. 4), letterpress inks

and the newsprint/magazine inks are known to differ in composition and polarity [3,16–19]. The presence of long-chain aliphatic hydrocarbons present as vehicles in some letterpress inks and in non-polar soils increases the phase inversion temperature. Short-chain aromatics and polar additives such as those present in many inks decrease the phase inversion temperature. These differences could account for the different relationship of surfactant cloud point and the process temperature for the optimum de-inking of different types of recovered paper.

2.1. One-dimensional nuclear magnetic resonance imaging studies

Optical techniques for studying the detachment of stains from fibers are limited by sample turbidity above the surfactant cloud point. Thus traditional light microscopy techniques cannot be used to study ink removal from fibers in pulp slurries or to study detergency near the surfactant cloud point. One-dimensional nuclear magnetic resonance (¹H NMR) imaging studies are a means of visualizing polar and non-polar soil and ink removal. This technique enables one to quantify removal rates from both fabric and paper substrates in the presence of surfactants having different cloud points.

The experimental design is described in Refs. 42 and 43. A circular piece of fabric or cloth 1 cm in diameter by 2 mm thick is glued to the bottom of

a glass sample tube 1 in in diameter by 1.25 in in length. Four drops of epoxy glue were used to attach the fabric or paper to the glass. The fibers were then saturated with about 50 mg of an oil phase. Dodecane was used because it is a reasonable model for the mineral oil commonly used as an ink vehicle in letterpress inks. $C_{12}(EO)_3$ and $C_{12}(EO)_4$ contained the indicated discrete number of carbon atoms in the hydrophobe and ethoxy (EO) groups in the hydrophile. $C_{12-15}(EO)_9$ is a commercial product containing 12–15 carbon atoms in the hydrophobe. This surfactant contains an average of nine EO group units with a roughly Poisson distribution of EO group chain lengths.

One-dimensional nuclear magnetic resonance (NMR) imaging studies indicate that maximum dodecane removal from polyester/cotton and cotton fabric was more rapid using $C_{12}(EO)_4$ than with $C_{12}(EO)_3$ [42,43]. The $C_{12}(EO)_4$ /dodecane phase inversion temperature (PIT) was approximately the same as the test temperatures, so the maximum soil removal rate occurred at approximately the phase inversion temperature. In contrast, the most active alcohol ethoxylate in removing dodecane from paper was the much more polar $C_{12-15}(EO)_9$. The $C_{12-15}(EO)_9$ PIT is considerable above the test temperature.

Even for $C_{12}(EO)_9$, dodecane removal was considerable slower from paper than from the cloth fabrics. For example, a nearly 100% dodecane removal from polyester/cotton was obtained in about 20 min with $C_{12}(EO)_4$. In contrast, $C_{12-15}(EO)_9$ removed only about 20% of the dodecane from paper during the same time period. After about 2 h, more than 60% of the dodecane remained in the paper.

These NMR results suggest an important effect of the substrate, paper fiber, on dodecane removal. The paper fibers were not disaggregated in the NMR experiment. Therefore it is difficult to interpret this effect in terms of de-inking phenomena. However, the higher dodecane detachment rate of $C_{12-15}(EO)_9$ in the NMR experiment is consistent with the known effects of surfactant polarity on wash de-inking effectiveness using mineral-oil-based inks. Maximum non-ionic de-inking surfactant effectiveness was observed at an HLB of

14.5–15.5 [19]. The HLB of $C_{12}(EO)_9$ is 13.3 while the HLB value of $C_{12}(EO)_4$ is 11.0.

2.2. Effect of pH on de-inking

Fiber swelling is promoted by high pH. This swelling weakens the attachment of ink to fibers, thereby aiding ink detachment during repulping. However, recent results [44] indicate that a high pulping pH is not essential to effective newsprint de-inking. Laboratory experiments indicated that reducing the pH to 5.5 and 3.5 did not reduce the de-inked sheet brightness or increase the residual ink particle content as indicated by image analysis.

2.3. Effect of mechanical forces on ink removal

The 1H NMR imaging studies were performed under static conditions. In addition, the paper fibers were not disaggregated, a process that occurs in repulping used paper. Figures 2 and 3 indicate that a printed character is in contact with many cellulose fibers. The disaggregation of these fibers can promote the break-up of printed characters into smaller ink particles and aid in ink detachment from fibers. Ink detachment is promoted by the purely mechanical energy forces of hydrodynamic flow of the aqueous phase, fiber swelling, fiber flexing, and fiber–fiber abrasion.

Both fiber–fiber abrasion and ink detachment increase with increasing pulping consistency. (Consistency is the content of paper in the aqueous slurry.) As noted above, separate process units are sometimes employed after pulping to promote ink detachment by fiber–fiber abrasion [4–13].

Fiber flexing can be increased by increasing the pulping consistency. This is because fiber flexing can accompany fiber–fiber collisions. Greater turbulence during pulping could also increase fiber flexing.

Mechanical forces are difficult to study quantitatively. They can sometimes be studied qualitatively in laboratory experiments by varying the rotor speed and rotor blade design used in repulping. Mechanical forces can mask the effect of chemical forces. For example, the effect of the surfactant cloud point on de-inking was discussed above. Strong mechanical forces could provide sufficient

ink removal to flatten the curves of Figs. 4 and 5, reducing the dependence of ink removal on the surfactant cloud point. Unpublished results from our laboratory indicate that it is possible to observe such effects. These effects are influenced by the ink chemistry.

The interaction of mechanical forces with chemical forces and their effect on ink detachment mechanisms are commented upon below.

3. Ink detachment from fibers

3.1. Liquid inks and soils

In laundering, the mechanism of soil removal from cloth fibers is dependent on the soil type, solid or liquid. Therefore, it seems reasonable that the mechanism of ink removal is dependent on the ink type. Liquid inks include most newsprint inks. Solid inks include photocopier and laser-printer inks (Fig. 3).

The shape of a liquid soil droplet or film can change during soil removal, permitting the surfactant to occupy the vacated space. In laundering, the primary mechanism of liquid soil removal from cotton fabric is the rollback mechanism (Fig. 6) [37,45,46]. Ink rollback may occur with liquid inks such as most newsprint inks. The aqueous surfactant solution preferentially wets the ink film on the paper surface. As a result, the film rolls back to form droplets (Fig. 6).

Figure 6 depicts 100% ink removal. In reality, some soil may be left behind. This marginal rollback has been proposed to occur in the laundering of oily soils [47]. Hydrodynamic forces during repulping promote the removal of these droplets

from cellulose surfaces. In a marginal rollback mechanism, these forces can affect the size of the droplet detached from the fiber surface.

Solid ink pigment particles may be carried away in the liquid droplets. Alternatively, lacking the liquid film that attaches them to the cellulose, the pigment particles may be independently separated from the cellulose by the mechanical action of the pulper.

Surfactants, by reducing the aqueous phase/oil interfacial tension, can penetrate liquid soils, causing them to disintegrate. Much of the oil phase is transferred to the interior of micelles in a process called solubilization [48]. Interfacial tension studies with model inks indicate that substantial reduction of the interfacial tension can occur during the same time frame as that required for the pulping process [43]. This indicates that, if the de-inking surfactant is above its CMC, ink solubilization can occur in the pulper. Solubilization (Fig. 7) consists of the following steps:

- (1) diffusion of surfactant micelles to the ink or soil surface;
- (2) adsorption of the micelle at the ink/water or soil/water interfaces;
- (3) the ink or soil species mix with the surfactant molecules;
- (4) the micelle desorbs from the surface with ink or soil in the interior of the micelle;
- (5) the micelle diffuses into the bulk of the solution.

Steps (1) and (5) can be promoted by hydrodynamic flow of the aqueous phase during repulping.

Emulsification is the breaking up of a large oil mass into small droplets between 0.1 and 10 μm in

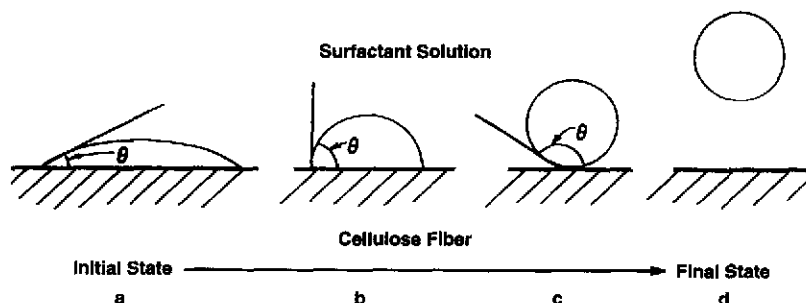


Fig. 6. Detachment of ink from cellulose by the rolling-up mechanism (adapted from Ref. 45, p. 232).

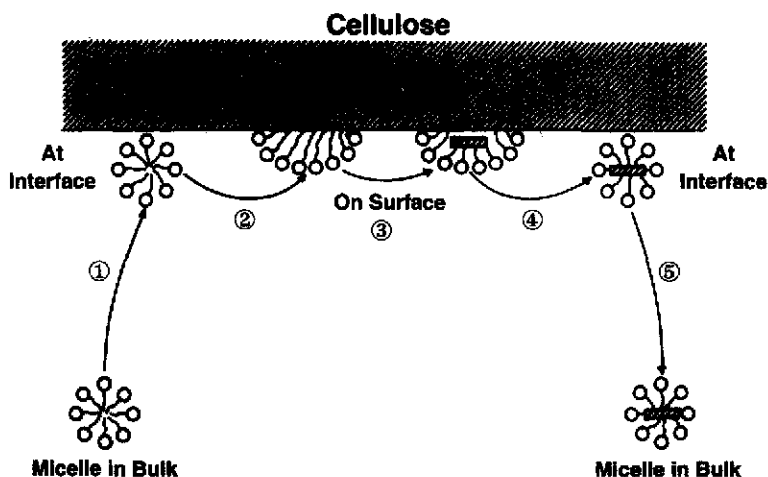


Fig. 7. Detachment of ink from cellulose by the solubilization mechanism (adapted from Ref. 45, p. 263).

diameter. Hydrodynamic forces during repulping can promote this break-up process. Emulsification can be promoted by low interfacial tension between the aqueous and the oil phases. Both strong stirring and a surfactant are required to prepare aqueous dispersions of oil-based inks [49]. These inks include letterpress and offset newsprint inks. In contrast, water-based ink such as flexographic newsprint ink are readily dispersed by stirring alone. Videogoniometer studies have demonstrated emulsification in the detachment of liquid soil from a single fiber [37].

Emulsification is favored by the presence of significant amounts of fatty acids, fatty alcohols, and other polar ingredients in the oil phase. Many ink and soil compositions contain significant amounts of such polar additives [3,16–19]. Flexographic newsprint inks contain water-soluble or dispersible acrylic binder resins. Ionization of these resins at the basic pH values typically used in repulping promotes the formation of very small ink particles (Table 3) [50].

The effect of aqueous phase pH on the zeta potential of ink dispersions and ink removal from these dispersions by flotation have been studied [51]. Reducing the pH of flexographic ink dispersions resulted in a decreased zeta potential and increased ink removal in flotation. The addition of dissolved calcium ions increased the ink particle size to about 60 μm , a particle size more efficiently

removed by flotation (Fig. 1) than the very small ink particles typical of flexographic ink dispersions (Table 3).

3.2. Solid inks

Electrical [20] and mechanical forces are particularly important for the adhesion of solid inks and soils to fibers. Solid inks include photocopier and laser-printer inks (Fig. 3). Morsink and Daane concluded that laser and xerographic inks behave like a solid soil since they are best detached from cellulose by anionic, not non-ionic, surfactants [52]. After de-inking, residual styrene-acrylate toner particles exhibit long indentations on their surfaces. These indentations were formerly occupied by cellulose fibers (Fig. 8). The survival of these indentations during pulping is consistent with the ink behaving like a solid soil.

Studies suggest that de-inking chemicals can reduce the softening point of photocopier and laser-printer inks sufficiently for the inks to have some liquid soil character [22]. At least some detached polyester-based toner ink particles exhibit smooth surfaces, suggesting that softening and reconfiguration of the ink particles had occurred (Fig. 9). Residual ink particles found after de-inking a mixed office waste containing about 80% laser-printed and photocopier paper also suggested that ink softening had occurred (Fig. 10).

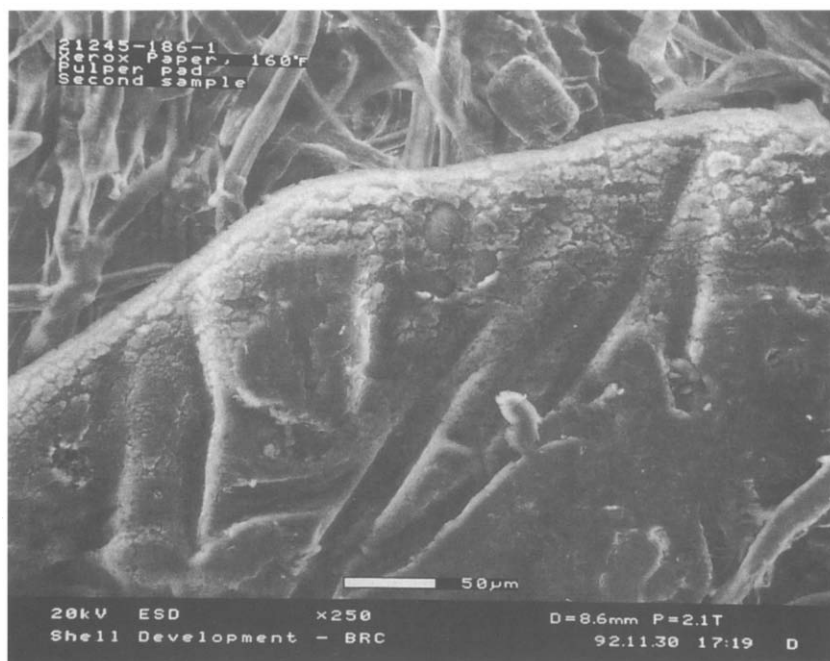


Fig. 8. Environmental scanning electron microscope image of styrene-acrylate copolymer toner ink particle after repulping photocopier-printed paper at 71°C.

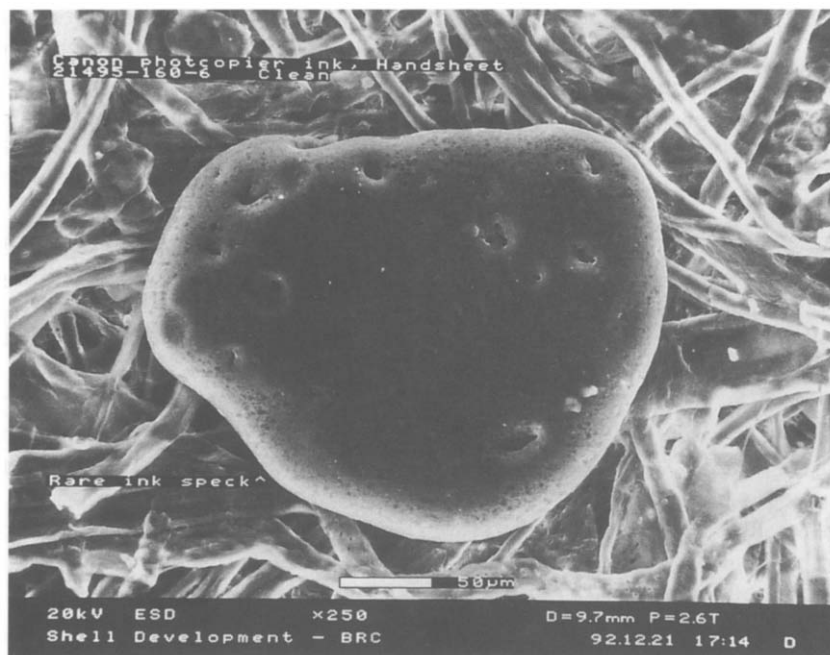


Fig. 9. Environmental scanning electron microscope image of a residual polyester toner ink particle after flotation of repulped photocopier-printed paper.



Fig. 10. Environmental scanning electron microscope image of a residual ink particle in de-inked office paper. This ink was processed in a pilot scale de-inking mill.

The geometry of the ink particles suggested that shear forces during repulping had elongated these particles. Environmental scanning electron microscopy and energy-dispersive X-ray analysis indicated that some of these residual ink particles (Fig. 11) resulted from the fusion of a toner ink particle containing dispersed iron mineral particles with another ink particle that did not contain any iron compounds. This fusion is promoted by both ink softening and mechanical forces during the particle–particle collision.

Two mechanisms are thought to operate in solid soil removal from fibers: wetting of the soil or fabric and adsorption of the surfactant at the fiber/liquid and soil/liquid interfaces [53]. For solid inks, the analogy to the first mechanism would be wetting of the paper and the ink by the surfactant solution. This reduces the adhesion of the ink particles to the cellulose. Cellulose swelling promoted by the high pulper pH also decreases the ink–cellulose adhesion forces. The mechanical action of the pulper assists in ink detachment from the cellulose.

The analogy to the second process is the adsorp-

tion of surfactant at the paper/surfactant solution and the ink/surfactant solution interfaces. This adsorption decreases the interfacial tension between the fiber and the aqueous phase and between the particles and the aqueous fluid, hence reducing particle adhesion resulting in less work required to detach the ink from the fiber. Again, mechanical action and high pH during pulping would assist the ink detachment process.

Contact angle studies indicate significant interactions between surfactants and solid inks [54]. The contact angle between a non-ionic surfactant solution and a laser-printer ink has been shown to change with time. Owing to the large surfactant solution volume, any changes in the contact angle probably reflect apparent surface energy changes of the ink.

3.3. Final comments

Direct parallels between de-inking and laundering are complicated by the greater role of mechanical forces in the de-inking processes. Mechanical disaggregation of paper in particular promotes ink



Fig. 11. Environmental scanning electron microscope image of two ink particles fused together. One contained high levels of iron, while the other particle contained little or no iron. The paper was a mixed office waste paper de-inked in a pilot scale paper mill.

detachment. The greater the role of mechanical forces in de-inking, the less influence chemical factors such as the surfactant cloud point have on ink removal efficiency. However, some parallels with laundering appear clear. Laboratory experiments indicate that ink removal from ledger paper, newspaper [40] and a newspaper/magazine mixture [41] is related to the de-inking surfactant cloud point — a clear parallel with detergency. This suggests that there are similarities in the mechanisms of ink and soil detachment from fibers.

Both the rollback mechanism and emulsification probably occur in the removal of liquid inks from cellulose fibers. The role of the solubilization mechanism in detaching ink from cellulose is less clear. This mechanism probably competes with ink detachment promoted by the mechanical disaggregation of fibers.

The removal of solid inks from cellulose is a complicated process. Softening of the ink facilitates ink detachment by hydrodynamic forces. This process may well dominate the surfactant-promoted wetting and adsorption effects thought to operate in solid soil removal from cloth [53]. Ink softening

can also promote the agglomeration of dispersed ink particles when they collide.

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